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Electronic device and control method therefor

Abstract

According to an embodiment, an electronic device comprises: a moving member comprising a motor, an operation performance member including an arm, a communication module comprising communication circuitry, a camera and at least one processor operatively connected to the moving member, the operation performance member, the communication module and the camera, wherein one or more of the at least one processor is configured to: control the moving member so that the electronic device moves to a position related to operation performance confirmed based on an operation performance command based on the operation performance command being received through the communication module, acquire a first marker image through the camera based on moving to the position, set a reference coordinate system based on the first marker image, set an operation area boundary within the reference coordinate system based on operation area boundary information acquired through a second marker image based on the second marker image being acquired through the camera, and control at least one of the operation performance members so that the same operates within the operation area boundary.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of International Application No. PCT/KR2022/004907 designating the United States, filed on Apr. 6, 2022, in the Korean Intellectual Property Receiving Office and claiming priority to Korean Patent Application No. 10-2021-0060157, filed on May 10, 2021, in the Korean Intellectual Property Office, the disclosures of each of which are incorporated by reference herein in their entireties.

BACKGROUND

Field

(1) The disclosure relates to an electronic device for setting a task area boundary and a control method thereof.

DESCRIPTION OF RELATED ART

(2) In the conventional industry, robot arms are typically fixed in a safe fence while being used, and the worker, facilities, and materials are protected from sudden movements of the robot arm due to the worker's mistakes or program errors in the fence by restricting the movable range of the robot arm or task area by program settings.

(3) For example, the task area of the robot arm is set based on the coordinate system (e.g., tool center position (TCP)) of the robot.

(4) Meanwhile, mobile robots have the advantage of being able to work in various spaces by combining a robot arm on top of a movable robot.

(5) Unlike the fixed robot arm, however, difficulty exists in restricting the task area of the robot arm for safety purposes as the shape or size of the task space, size, content of task, or task reference coordinate system information vary whenever it moves.

(6) The arm of the mobile robot (or electronic device) should process the task fitting the task space while moving various task spaces.

(7) When processing a task, the arm of the mobile robot docks the robot in a predetermined place and performs the assigned task. Upon docking, a tiny error occurs in the robot docking position, causing a failure in achieving the task with respect to the robot coordinate system.

(8) Due to such issue, after docking in the task space, the mobile robot arm uses a separate reference coordinate system for the task assigned by recognition of the markers attached in the task place, such as a jig or facility through a camera attached to the robot arm.

(9) However, since it obtains the reference coordinate system through the attached markers in every task space to perform the task, it may not use robot task area restriction information based on the robot coordinate system.

SUMMARY

(10) Embodiments of the disclosure provide an electronic device and a control method that sets a robot task area boundary upon robot working in various spaces.

(11) According to various example embodiments, an electronic device may comprise: a moving member comprising a motor, a task performing member comprising an articulated arm, a communication module comprising communication circuitry, a camera, and at least one processor operatively connected to the moving member, the task performing member, the communication

module, and the camera. One or more of the at least one processor may be configured to: upon receiving a task performance command through the communication module, control the moving member to move to a task performing-related position identified based on the task performance command, obtain a first marker image through the camera based on moving to the position, set a reference coordinate system based on the first marker image, upon obtaining a second marker image through the camera, set a task area boundary in the reference coordinate system based on task area boundary information obtained through the second marker image, and control at least one task performing member to operate within the task area boundary.

(12) According to various example embodiments, a method for controlling an electronic device may comprise, upon receiving a task performance command moving to a task performing-related position identified based on the task performance command, obtaining a first marker image through a camera based on moving to the position, setting a reference coordinate system based on the first marker image, upon obtaining a second marker image through the camera, setting a task area boundary in the reference coordinate system based on task area boundary information obtained through the second marker image, and controlling the electronic device to operate within the task area boundary.

(13) According to various example embodiments, a display device may comprise: a camera, a display, and at least one processor operatively connected to the camera and the display. One or more of the at least one processor may be configured to: display, on the display, an image for a task space including a first marker image through the camera, receive a drawing input related to a task area boundary in the task space displayed on the display, display, in augmented reality, the drawing input in the task space displayed on the display, generate task area boundary information based on the drawing input on a reference coordinate system set based on the first marker image, and generate a second marker image including the task area boundary information.

(14) An electronic device according to various example embodiments of the disclosure may set a task area boundary that is not affected by an error that occurs upon docking in a task space.

(15) Further, since it is not required to store information about a plurality of task places in the electronic device, the system design may be easier.

(16) Further, upon designing a task place, the user may preset a task area boundary.

(17) It is also possible to intuitively provide the user with a task boundary area through augmented reality technology.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above and other aspects, features and advantages of certain embodiments of the present disclosure will be more apparent from the following detailed description, taken in conjunction with the accompanying drawings, in which:

(2) FIG. 1 is a block diagram illustrating an example electronic device in a network environment according to various embodiments;

(3) FIG. 2 is a diagram illustrating an example configuration of a mobile robot-type electronic device according to various embodiments;

(4) FIG. 3 is a flowchart illustrating an example operation of setting a task area boundary by an electronic device according to various embodiments;

(5) FIG. 4 is a diagram illustrating an example operation of setting a task area boundary surface by an electronic device according to various embodiments;

(6) FIG. 5A is a diagram illustrating an example first marker image and an example second marker image according to various embodiments;

(7) FIG. 5B is a diagram illustrating an example first marker image and an example second marker

image according to various embodiments;

(8) FIG. 5C is a diagram illustrating an example first marker image and an example second marker image according to various embodiments;

(9) FIG. 5D is a diagram illustrating an example beacon device for task area boundary information and an example first marker image according to various embodiments;

(10) FIG. 6 is a diagram illustrating a polygonal task area boundary of an electronic device according to various embodiments;

(11) FIG. 7 is a diagram illustrating a hexagonal task area boundary of an electronic device according to various embodiments;

(12) FIG. 8 is a diagram illustrating a movable area for each axis of an electronic device according to various embodiments;

(13) FIG. 9 is a diagram illustrating an example operation of displaying a task area boundary in augmented reality according to various embodiments; and

(14) FIG. 10 is a flowchart illustrating an example operation of generating a marker image including task area boundary information using augmented reality according to various embodiments.

DETAILED DESCRIPTION

(15) FIG. 1 is a block diagram illustrating an example electronic device **101** in a network environment **100** according to various embodiments. Referring to FIG. 1, the electronic device **101** in the network environment **100** may communicate with at least one of an electronic device **102** via a first network **198** (e.g., a short-range wireless communication network), or an electronic device **104** or a server **108** via a second network **199** (e.g., a long-range wireless communication network). According to an embodiment, the electronic device **101** may communicate with the electronic device **104** via the server **108**. According to an embodiment, the electronic device **101** may include a processor **120**, memory **130**, an input module **150**, a sound output module **155**, a display module **160**, an audio module **170**, a sensor module **176**, an interface **177**, a connecting terminal. According to an embodiment, the display module **160** may include a first display module **351** corresponding to the user's left eye and/or a second display module **353** corresponding to the user's right eye, a haptic module **179**, a camera module **180**, a power management module **188**, a battery **189**, a communication module **190**, a subscriber identification module (SIM) **196**, or an antenna module **197**. In an embodiment, at least one (e.g., the connecting terminal **178**) of the components may be omitted from the electronic device **101**, or one or more other components may be added in the electronic device **101**. According to an embodiment, some (e.g., the sensor module **176**, the camera module **180**, or the antenna module **197**) of the components may be integrated into a single component (e.g., the display module **160**).

(16) The processor **120** may execute, for example, software (e.g., a program **140**) to control at least one other component (e.g., a hardware or software component) of the electronic device **101** coupled with the processor **120**, and may perform various data processing or computation. According to an embodiment, as at least part of the data processing or computation, the processor **120** may store a command or data received from another component (e.g., the sensor module **176** or the communication module **190**) in volatile memory **132**, process the command or the data stored in the volatile memory **132**, and store resulting data in non-volatile memory **134**. According to an embodiment, the processor **120** may include a main processor **121** (e.g., a central processing unit (CPU) or an application processor (AP)), or an auxiliary processor **123** (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor **121**. For example, when the electronic device **101** includes the main processor **121** and the auxiliary processor **123**, the auxiliary processor **123** may be configured to use lower power than the main processor **121** or to be specified for a designated function. The auxiliary processor **123** may be implemented as separate from, or as part of the main

processor **121**.

(17) The auxiliary processor **123** may control at least some of functions or states related to at least one component (e.g., the display module **160**, the sensor module **176**, or the communication module **190**) among the components of the electronic device **101**, instead of the main processor **121** while the main processor **121** is in an inactive (e.g., sleep) state, or together with the main processor **121** while the main processor **121** is in an active state (e.g., executing an application). According to an embodiment, the auxiliary processor **123** (e.g., an image signal processor or a communication processor) may be implemented as part of another component (e.g., the camera module **180** or the communication module **190**) functionally related to the auxiliary processor **123**. According to an embodiment, the auxiliary processor **123** (e.g., the neural processing unit) may include a hardware structure specified for artificial intelligence model processing. The artificial intelligence model may be generated via machine learning. Such learning may be performed, e.g., by the electronic device **101** where the artificial intelligence is performed or via a separate server (e.g., the server **108**). Learning algorithms may include, but are not limited to, e.g., supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning. The artificial intelligence model may include a plurality of artificial neural network layers. The artificial neural network may be a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted Boltzmann machine (RBM), a deep belief network (DBN), a bidirectional recurrent deep neural network (BRDNN), deep Q-network or a combination of two or more thereof but is not limited thereto. The artificial intelligence model may, additionally or alternatively, include a software structure other than the hardware structure.

(18) The memory **130** may store various data used by at least one component (e.g., the processor **120** or the sensor module **176**) of the electronic device **101**. The various data may include, for example, software (e.g., the program **140**) and input data or output data for a command related thereto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**.

(19) The program **140** may be stored in the memory **130** as software, and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.

(20) The input module **150** may receive a command or data to be used by other component (e.g., the processor **120**) of the electronic device **101**, from the outside (e.g., a user) of the electronic device **101**. The input module **150** may include, for example, a microphone, a mouse, a keyboard, keys (e.g., buttons), or a digital pen (e.g., a stylus pen).

(21) The sound output module **155** may output sound signals to the outside of the electronic device **101**. The sound output module **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record. The receiver may be used for receiving incoming calls. According to an embodiment, the receiver may be implemented as separate from, or as part of the speaker.

(22) The display module **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display **160** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to an embodiment, the display **160** may include a touch sensor configured to detect a touch, or a pressure sensor configured to measure the intensity of a force generated by the touch.

(23) The audio module **170** may convert a sound into an electrical signal and vice versa. According to an embodiment, the audio module **170** may obtain the sound via the input module **150**, or output the sound via the sound output module **155** or a headphone of an external electronic device (e.g., an electronic device **102**) directly (e.g., wiredly) or wirelessly coupled with the electronic device **101**.

(24) The sensor module **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and then generate an electrical signal or data value corresponding to the detected state. According to an embodiment, the sensor module **176** may include, for example, a gesture sensor, a

gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an accelerometer, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.

(25) The interface **177** may support one or more specified protocols to be used for the electronic device **101** to be coupled with the external electronic device (e.g., the electronic device **102**) directly (e.g., wiredly) or wirelessly. According to an embodiment, the interface **177** may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.

(26) A connecting terminal **178** may include a connector via which the electronic device **101** may be physically connected with the external electronic device (e.g., the electronic device **102**). According to an embodiment, the connecting terminal **178** may include, for example, a HDMI connector, a USB connector, a SD card connector, or an audio connector (e.g., a headphone connector).

(27) The haptic module **179** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or motion) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to an embodiment, the haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.

(28) The camera module **180** may capture a still image or moving images. According to an embodiment, the camera module **180** may include one or more lenses, image sensors, image signal processors, or flashes.

(29) The power management module **188** may manage power supplied to the electronic device **101**. According to an embodiment, the power management module **188** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

(30) The battery **189** may supply power to at least one component of the electronic device **101**. According to an embodiment, the battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.

(31) The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101** and the external electronic device (e.g., the electronic device **102**, the electronic device **104**, or the server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more communication processors that are operable independently from the processor **120** (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment, the communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device **104** via a first network **198** (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or a second network **199** (e.g., a long-range communication network, such as a legacy cellular network, a 5G network, a next-generation communication network, the Internet, or a computer network (e.g., local area network (LAN) or wide area network (WAN))). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **192** may identify or authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the subscriber identification module **196**.

(32) The wireless communication module **192** may support a 5G network, after a 4G network, and next-generation communication technology, e.g., new radio (NR) access technology. The NR

access technology may support enhanced mobile broadband (eMBB), massive machine type communications (mMTC), or ultra-reliable and low-latency communications (URLLC). The wireless communication module **192** may support a high-frequency band (e.g., the mmWave band) to achieve, e.g., a high data transmission rate. The wireless communication module **192** may support various technologies for securing performance on a high-frequency band, such as, e.g., beamforming, massive multiple-input and multiple-output (massive MIMO), full dimensional MIMO (FD-MIMO), array antenna, analog beam-forming, or large scale antenna. The wireless communication module **192** may support various requirements specified in the electronic device **101**, an external electronic device (e.g., the electronic device **104**), or a network system (e.g., the second network **199**). According to an embodiment, the wireless communication module **192** may support a peak data rate (e.g., 20 Gbps or more) for implementing eMBB, loss coverage (e.g., 164 dB or less) for implementing mMTC, or U-plane latency (e.g., 0.5 ms or less for each of downlink (DL) and uplink (UL), or a round trip of 1 ms or less) for implementing URLLC.

(33) The antenna module **197** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device). According to an embodiment, the antenna module **197** may include an antenna including a radiator formed of a conductive body or conductive pattern formed on a substrate (e.g., a printed circuit board (PCB)). According to an embodiment, the antenna module **197** may include a plurality of antennas (e.g., an antenna array). In this case, at least one antenna appropriate for a communication scheme used in a communication network, such as the first network **198** or the second network **199**, may be selected from the plurality of antennas by, e.g., the communication module **190**. The signal or the power may then be transmitted or received between the communication module **190** and the external electronic device via the selected at least one antenna. According to an embodiment, other parts (e.g., radio frequency integrated circuit (RFIC)) than the radiator may be further formed as part of the antenna module **197**.

(34) According to various embodiments, the antenna module **197** may form a mmWave antenna module. According to an embodiment, the mmWave antenna module may include a printed circuit board, a RFIC disposed on a first surface (e.g., the bottom surface) of the printed circuit board, or adjacent to the first surface and capable of supporting a designated high-frequency band (e.g., the mmWave band), and a plurality of antennas (e.g., array antennas) disposed on a second surface (e.g., the top or a side surface) of the printed circuit board, or adjacent to the second surface and capable of transmitting or receiving signals of the designated high-frequency band.

(35) At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

(36) According to an embodiment, commands or data may be transmitted or received between the electronic device **101** and the external electronic device **104** via the server **108** coupled with the second network **199**. The external electronic devices **102** or **104** each may be a device of the same or a different type from the electronic device **101**. According to an embodiment, all or some of operations to be executed at the electronic device **101** may be executed at one or more of the external electronic devices **102**, **104**, or **108**. For example, if the electronic device **101** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device **101**. The electronic device **101** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example.

The electronic device **101** may provide ultra low-latency services using, e.g., distributed computing or mobile edge computing. In an embodiment, the external electronic device **104** may include an Internet-of-things (IoT) device. The server **108** may be an intelligent server using machine learning and/or a neural network. According to an embodiment, the external electronic device **104** or the server **108** may be included in the second network **199**. The electronic device **101** may be applied to intelligent services (e.g., smart home, smart city, smart car, or health-care) based on 5G communication technology or IoT-related technology.

(37) FIG. 2 is a diagram illustrating an example configuration of a mobile robot-type electronic device according to various embodiments.

(38) According to various embodiments, referring to FIG. 2, an electronic device **101** (e.g., the electronic device **101** of FIG. 1) includes a moving member (e.g., including a motor) **210**, a task performing member (e.g., including an arm) **220**, a camera **230** (e.g., the camera module **180** of FIG. 1), and a processor (e.g., including processing circuitry) **240** (e.g., the processor **120** of FIG. 1).

(39) According to various embodiments, the electronic device **101** may perform a task suitable for the corresponding task space while moving the task space. For example, the electronic device **101** may include a mobile robot.

(40) According to various embodiments, the moving member **210** may move the position of the electronic device **101** under the control of the processor **240**. For example, the moving part **210** may include at least one of at least one wheel, a roller, a belt, or the like that may be driven by, for example, and without limitation, a motor, and may move on the bottom surface of the task space or through a rail installed on the bottom of the task space.

(41) According to various embodiments, the task performing member **220** may include a component for performing a corresponding task in the task space under the control of the processor **240**, and may include, for example, and without limitation, a robot arm. For example, the task performing member **220** may include a six-axis articulated robot arm.

(42) According to various embodiments, the camera **230** may capture at least one of the first marker image or the second marker image attached to the task space. For example, the first marker image may include information related to the reference coordinate system, and the second marker image may include task area boundary information. For example, the task area boundary information may include at least one of coordinate information about at least one boundary surface, a slope, a shape of the task area boundary, or a minimum or maximum moving angle for each axis of the task performing member.

(43) According to various embodiments, the processor **240** may include various processing circuitry (as used herein, including the claims, the term “processor” may include various processing circuitry, including at least one processor, wherein one or more processors of the at least one processor may be configured to perform the various functions described herein) and set the reference coordinate system of the task space based on the information related to the reference coordinate system.

(44) According to various embodiments, one or more of the processor **240** may set a task area boundary in the reference coordinate system based on the task boundary information.

(45) An operation of setting a task area boundary of the electronic device **101** according to various embodiments is described in greater detail below with reference to FIGS. 3 to 8.

(46) FIG. 3 is a flowchart illustrating an example operation of setting a task area boundary by an electronic device according to various embodiments.

(47) According to various embodiments, in operation **310**, if a task command is received, the electronic device (e.g., the electronic device **101** of FIG. 1, the processor **120** of FIG. 1, the electronic device **101** of FIG. 2, or the processor **240** of FIG. 2) may move to a position related to task performing, identified based on the task performance command.

(48) For example, when a task command is received from an external electronic device through a

communication module (e.g., the communication module **190** of FIG. **1**), the electronic device may move to the corresponding position. According to various embodiments, the electronic device may move to the corresponding position and dock at the corresponding position.

(49) According to various embodiments, in operation **320**, the electronic device may obtain a first marker image through a camera (e.g., the camera module **180** of FIG. **1** or the camera **230** of FIG. **2**) after (e.g., based on) moving to the position. For example, the first marker image may include reference coordinate system information for setting a reference coordinate system. According to various embodiments, the first marker image may include a checker, a barcode, or a QR code.

(50) According to various embodiments, in operation **330**, the electronic device may configure (e.g., set) the reference coordinate system based on the first marker image.

(51) For example, the electronic device may obtain reference coordinate system information from the first marker image and may set the reference coordinate system based on the obtained reference coordinate system information. For example, the electronic device may set a reference coordinate system including an x-axis, a y-axis, and a z-axis, with the marker position as the origin of the reference coordinate system.

(52) According to various embodiments, when the electronic device fails to set the reference coordinate system, the electronic device may re-obtain the first marker image through the camera and re-set the reference coordinate system.

(53) According to various embodiments, in operation **340**, upon obtaining a second marker image through the camera, the electronic device may set a task area boundary in the reference coordinate system based on task area boundary information obtained through the second marker image. For example, the second marker image may include task area boundary information. According to various embodiments, the second marker image may include, for example, and without limitation, a checker, a barcode, a QR code, or the like.

(54) According to various embodiments, the second marker image may be generated based on a user's drawing input related to the task area boundary, input to an external display device. The operation of generating the second marker image according to various embodiments is described in greater detail below with reference to FIGS. **9** and **10**.

(55) According to various embodiments, the second marker image may be disposed adjacent to the first marker image or may be disposed in the first marker image. The arrangement of the first marker image and the second marker image according to various embodiments is described in greater detail below with reference to FIGS. **5A**, **5B** and **5C**.

(56) FIG. **5A** is a diagram illustrating an example first marker image and an example second marker image according to various embodiments.

(57) According to various embodiments, referring to FIG. **5A**, the first marker image may be a checker (e.g., checkered pattern, checkerboard, etc.), the second marker image may be a QR code, and the second marker image may be disposed adjacent to one side of the first marker image. Although FIG. **5A** illustrates that the second marker image is disposed on the right side of the first marker image, the disclosure is not limited thereto, and the second marker image may be disposed on the left side, the upper side, or the lower side of the first marker image.

(58) FIG. **5B** is a diagram illustrating an example first marker image and an example second marker image according to various embodiments.

(59) According to various embodiments, referring to FIG. **5B**, the first marker image may be a checker (e.g., including a checkered pattern, checkerboard, etc.), the second marker image may be a barcode, and the second marker image may be disposed adjacent to one side of the first marker image. Although FIG. **5B** illustrates that the second marker image is disposed on the upper side of the first marker image, the disclosure is not limited thereto, and the second marker image may be disposed on the left side, the right side, or the lower side of the first marker image.

(60) FIG. **5C** is a diagram illustrating an example first marker image and an example second marker image according to various embodiments.

(61) According to various embodiments, referring to FIG. 5C, the first marker image may be a checker (e.g., including a checkered pattern, checkerboard, etc.), the second marker image may be a QR code, and the second marker image may be disposed in the first marker image.

(62) According to various embodiments, the electronic device may receive task area boundary information from a beacon device through the communication module. For example, the electronic device may receive task area boundary information through at least one of the second marker image or the beacon device.

(63) FIG. 5D is a diagram illustrating an example beacon device for task area boundary information and an example first marker image according to various embodiments.

(64) According to various embodiments, referring to FIG. 5D, the first marker image may be a checker (e.g., including a checkered pattern, a checkerboard, etc.), and the beacon device may be disposed adjacent to the first marker image. According to various embodiments, the beacon device may be disposed at a position capable of transmitting and receiving information to and from the electronic device regardless of the position of the first marker image.

(65) According to various embodiments, the electronic device may set the task area boundary in the reference coordinate system based on the task area boundary information received from the beacon device.

(66) According to various embodiments, the task area boundary information may include at least one of coordinate information about at least one boundary surface, a slope, a shape of the task area boundary, or a minimum or maximum moving angle for each axis of the task performing member (e.g., the task performing member **220** of FIG. 2).

(67) For example, the task area boundary information may include 6DOF (degrees of freedom) coordinate information such as (x, y, z, Ox, Oy, Oz). For example, x, y, and z may represent the position of the boundary surface with respect to the origin of the three-dimensional coordinate system, and Ox, Oy, and Oz may represent the angles at which the object is rotated counterclockwise about the respective axes (e.g., the x-axis, y-axis, and z-axis) as the slope of the boundary surface. An operation of setting a task area boundary based on task area boundary information according to various embodiments is described in greater detail below with reference to FIG. 4.

(68) FIG. 4 is a diagram illustrating an example operation of setting a task area boundary surface by an electronic device according to various embodiments.

(69) According to various embodiments, referring to FIG. 4, an electronic device **101** (e.g., the electronic device **101** of FIG. 1, the processor **120** of FIG. 1, the electronic device **101** of FIG. 2, or the processor **240** of FIG. 2) may obtain a first marker image **410** through a camera (e.g., the camera module **180** of FIG. 1 or the camera **230** of FIG. 2) and may set a reference coordinate system **411** based on reference coordinate system information included in the first marker image.

(70) According to various embodiments, the electronic device may obtain a second marker image **420** through the camera after setting the reference coordinate system **411**, and may set a task area boundary based on the task area boundary information included in the second marker image. For example, referring to FIG. 4, the electronic device **101** may set a boundary surface **421** based on one coordinate information included in the task area boundary information.

(71) For example, when the coordinate information is (0, 5, 0, 0, 0, 80), the boundary surface may be rotated by 80 degrees about the z-axis from a position 5 degrees away from the origin of the reference coordinate system in the y-axis direction.

(72) According to various embodiments, the shape of the task area boundary included in the task area boundary information may be obtained as the plurality of boundary surfaces cross each other. For example, the task area boundary information may include at least one of coordinate information or the slope of the plurality of boundary surfaces.

(73) According to various embodiments, the electronic device may obtain information about at least one crossing line where the plurality of boundary surfaces cross each other based on at least one of

the coordinate information or slope of the plurality of boundary surfaces and set a task area boundary based on the coordinate information, slope, and information about the cross line of the plurality of boundary surfaces. An operation of setting a task area boundary based on a plurality of boundary surfaces is described below with reference to FIG. 6.

(74) FIG. 6 is a diagram illustrating a polygonal task area boundary of an electronic device according to various embodiments.

(75) According to various embodiments, referring to FIG. 6, an electronic device (e.g., the electronic device **101** of FIG. 1, the processor **120** of FIG. 1, the electronic device **101** of FIG. 2, or the processor **240** of FIG. 2) may set a task area boundary **610** based on task area boundary information obtained from a second marker image in a reference coordinate system set based on a first marker image. According to various embodiments, the task area boundary information may be obtained from a beacon device.

(76) For example, the second marker image may include three pieces of 6DOF coordinate information, and the task area boundary **610** may include three boundary surfaces **611**, **612**, and **613** corresponding to the three pieces of 6DOF coordinate information.

(77) For example, the electronic device may set a first boundary surface **611** based on first coordinate information $0, P1, 0, 0, 0, \theta1$, set a second boundary surface **612** based on second coordinate information $0, P2, 0, 0, 0, \theta2$, and set a third boundary surface **613** based on third coordinate information $P3, 0, 0, 0, 0, 90^\circ-\theta3$.

(78) According to various embodiments, the electronic device may obtain information about a first crossing line where the first boundary surface **611** and the second boundary surface **612** cross each other, and may obtain information about a second crossing line where the second boundary surface **612** and the third boundary surface **613** cross each other.

(79) According to various embodiments, the electronic device may set a task boundary area **610** based on information about the first boundary surface **611** to the third boundary surface **613** and information about the first crossing line and the second crossing line in the reference coordinate system.

(80) According to various embodiments, information about the shape of the task area boundary included in the task area boundary information may be included. For example, the task area boundary information may include information about two points and information that the shape of the task area boundary is a hexahedral shape having two points as vertices. An operation of setting a task area boundary when information about the shape of a task area boundary included in task area boundary information is included according to various embodiments is described in greater detail below with reference to FIG. 7.

(81) FIG. 7 is a diagram illustrating a hexagonal task area boundary of an electronic device according to various embodiments.

(82) According to various embodiments, referring to FIG. 7, an electronic device **101** (e.g., the electronic device **101** of FIG. 1, the processor **120** of FIG. 1, the electronic device **101** of FIG. 2, or the processor **240** of FIG. 2) may set a task area boundary **720** based on task area boundary information obtained from a second marker image in a reference coordinate system **710** set based on a first marker image. According to various embodiments, the task area boundary information may be obtained from a beacon device.

(83) For example, the second marker image may include a plurality of coordinate information and information about the shape of the task area boundary. For example, the second marker image may include two pieces of coordinate information U and U' and information indicating that the task area boundary is a hexahedral.

(84) According to various embodiments, based on the task area boundary information included in the second marker image, a hexahedral task area boundary **720** having the first coordinate U and the second coordinate U' as diagonal vertices may be set.

(85) According to various embodiments, the task area boundary information may include a

minimum or maximum moving angle for each axis of the task performing member (e.g., the task performing member **220** of FIG. 2) in the task area boundary. According to various embodiments, the electronic device (e.g., the electronic device **101** of FIG. 1, the processor **120** of FIG. 1, the electronic device **101** of FIG. 2, or the processor **240** of FIG. 2) may limit the moving angle range of each axis of the task performing member, based on the minimum or maximum moving angle for each axis of the task performing member included in the task area boundary information. An operation of limiting the moving angle range of each axis of the task performing member according to various embodiments is described with reference to FIG. 8.

(86) FIG. 8 is a diagram illustrating an example movable area for each axis of an electronic device according to various embodiments.

(87) According to various embodiments, referring to FIG. 8, the electronic device (e.g., the electronic device **101** of FIG. 1, the processor **120** of FIG. 1, the electronic device **101** of FIG. 2, or the processor **240** of FIG. 2) may limit the moving angle range of each axis (e.g., axis 1, axis 2, axis 3, axis 4, axis 5 and axis 6) of the task performing member of the electronic device in the reference coordinate system.

(88) According to various embodiments, even if the task area boundary information does not include information about the minimum or maximum moving angle for each axis of the task performing member, the electronic device may set the moving angle range for each axis of the task performing member of the electronic device based on the task area boundary set in the reference coordinate system and the position of the electronic device.

(89) According to various embodiments, the moving angle range for each axis of the task performing member may be set to an absolute value range of the moving angle for each axis, or may be set in the form of an equation in which the moving angle range of the corresponding axis is changed according to the angle of another axis.

(90) According to various embodiments, if setting the task area boundary in the reference coordinate system fails, the electronic device (e.g., the electronic device **101** of FIG. 1, the processor **120** of FIG. 1, the electronic device **101** of FIG. 2, or the processor **240** of FIG. 2) may reobtain the second marker image through a camera (e.g., the camera module **180** of FIG. 1 or the camera **230** of FIG. 2). According to various embodiments, the electronic device may re-receive task area boundary information from a beacon device through a communication module (e.g., the communication module **190** of FIG. 1).

(91) Referring back to FIG. 3, according to various embodiments, in operation **350**, the electronic device may control at least one task performing member (e.g., the task performing member **220** of FIG. 2) to operate within a task boundary.

(92) According to various embodiments, the electronic device may transmit, to an external display device through the communication module, an image in which the task area boundary is displayed in augmented reality in the image obtained through the camera. For example, the electronic device may obtain an image (e.g., a still image or a video) captured for the task space through the camera, and may transmit an image in which the set task area boundary is displayed in augmented reality to the external display device. Accordingly, it is possible to intuitively display the task area boundary in the task space of the electronic device through the external display device and provide the same to the user.

(93) FIG. 9 is a diagram illustrating an example operation of displaying a task area boundary in augmented reality according to various embodiments.

(94) According to various embodiments, referring to FIG. 9, a display device **900** (e.g., the electronic device of FIG. 1) may display an image captured for a task space including a first marker image **910** for setting a reference coordinate system.

(95) According to various embodiments, the electronic device **101** (e.g., the electronic device **101** of FIG. 1 or the electronic device **101** of FIG. 2) may be disposed in the task space, and in a state in which the electronic device **101** is not disposed, an object corresponding to the electronic device

101 may be input to the image captured for the task space by a user input.

(96) According to various embodiments, the display device **900** may receive the task area boundary **920** of the electronic device **101** by, for example, a user input. For example, the display device **900** may receive the user's drawing input of drawing a task area boundary **920** through a touchscreen (e.g., the display module **160** of FIG. **1**). In an embodiment, the display device **900** may receive the user's drawing input, input through an external input device (e.g., a mouse or a drawing pad), from the external input device.

(97) According to various embodiments, the display device **900** may generate a marker image including task area boundary **920** information received by a user input. An operation of generating a marker image including task area boundary information using augmented reality is described in greater detail below with reference to FIG. **10**.

(98) FIG. **10** is a flowchart illustrating an example operation of generating a marker image including task area boundary information using augmented reality according to various embodiments.

(99) According to various embodiments, in operation **1001**, a display device (e.g., the electronic device **101** of FIG. **1** or the display device **900** of FIG. **9**) may execute an AR camera application.

(100) According to various embodiments, in operation **1002**, the display device may capture a task place and a marker. For example, the display device may capture the task place and the marker for setting the reference coordinate system using the camera (e.g., the camera module **180** of FIG. **1**).

(101) According to various embodiments, in operation **1003**, the display device may install a task boundary (e.g., an AR object) by anchoring the marker. For example, the display device may fix the position of the marker in the image, dispose a mobile robot (e.g., the electronic device **101** of FIG. **1** or the electronic device **101** of FIG. **2**) as an augmented reality (AR) object in the image, and dispose a task area boundary as an AR object. For example, one of a plurality of area boundary objects stored in an application may be selected and disposed, as the AR object of a task area boundary, or may be disposed through the user's drawing input.

(102) According to various embodiments, in operation **1004**, the display device may determine whether the task area boundary installation is successful. For example, it may be determined whether the installation of at least one of the augmented reality object of the mobile robot or the augmented reality object of the task area boundary in the image is successful. For example, when the augmented reality object of the mobile robot is installed in the air or the augmented reality object of the task area boundary is installed in a place irrelevant to the task area of the mobile robot, the display device may determine that the task area boundary installation fails.

(103) According to various embodiments, when it is determined that the installation of the task area boundary fails (no in operation **1004**), the display device may return to operation **1003** to reinstall the task area boundary.

(104) According to various embodiments, when it is determined that the installation of the task area boundary is successful (yes in operation **1004**), in operation **1005**, the display device may calculate the coordinates of the task boundary having the coordinates in the marker as the origin. For example, the display device may set the reference coordinate system by setting the marker for setting the reference coordinate system as the origin of the reference coordinate system, and calculate the coordinates of the task area boundary.

(105) According to various embodiments, in operation **1006**, the display device may check the validity of the calculated coordinates of the task area boundary. For example, the display device may determine whether the coordinates of the task area boundary are valid in relation to the possibility of actual utilization in consideration of the design of the actual task space and the position or size of the machine or article disposed in the actual task space.

(106) According to various embodiments, when it is determined that the task area boundary is not valid (no in operation **1006**), the display device may return to operation **1003** to reinstall the task area boundary.

(107) According to various embodiments, when it is determined that the task area boundary is valid (yes in operation **1006**), the display device may generate a QR code including the task area boundary information in operation **1007**. According to various embodiments, the display device may generate a barcode or checker including task area boundary information, or may transmit the task area boundary information to the beacon device.

(108) According to various embodiments, in operation **1008**, the display device may print the QR code. According to various embodiments, the printed QR code may be attached within the task area. For example, the printed QR code may be attached adjacent to or within the marker image for setting the reference coordinate system, as shown in FIGS. 5A to 5C.

(109) According to various example embodiments, an electronic device (e.g., the electronic device of FIG. 1 or the electronic device **101** of FIG. 2) may comprise: a moving member including a motor (e.g., the moving member **210** of FIG. 2), a task performing member comprising an arm (e.g., the task performing member **220** of FIG. 2), a communication module comprising communication circuitry (e.g., the communication module **190** of FIG. 1), a camera (e.g., the camera module **180** of FIG. 1 or the camera **230** of FIG. 20), and at least one processor (e.g., the processor **120** of FIG. 1 or the processor **240** of FIG. 2) operatively connected to the moving member, the task performing member, the communication module, and the camera. One or more of the at least one processor may be configured to: upon receiving a task performance command through the communication module, control the moving member to move to a task performing-related position identified based on the task performance command, obtain a first marker image through the camera based on moving to the position, set a reference coordinate system based on the first marker image, upon obtaining a second marker image through the camera, set a task area boundary in the reference coordinate system based on task area boundary information obtained through the second marker image, and control at least one task performing member to operate within the task area boundary.

(110) According to various example embodiments, the task area boundary information may include at least one of coordinate information about at least one boundary surface, a slope, a shape of the task area boundary, or a minimum or maximum moving angle for each axis of the task performing member.

(111) According to various example embodiments, the task area boundary information may include at least one of coordinate information about a plurality of boundary surfaces or a slope. One or more of the at least one processor may be configured to: obtain information about at least one crossing line where the plurality of boundary surfaces cross each other based on at least one of the coordinate information about the plurality of boundary surfaces or the slope and set the task area boundary based on the coordinate information about the plurality of boundary surfaces, the slope, and the information about the crossing line.

(112) According to various example embodiments, one or more of the at least one processor may be configured to, based on setting the reference coordinate system failing, reobtain the first marker image through the camera.

(113) According to various example embodiments, one or more of the at least one processor may be configured to, based on setting the task area boundary failing, reobtain the second marker image through the camera.

(114) According to various example embodiments, one or more of the at least one processor may be configured to transmit an image obtained by displaying, in augmented reality, the task area boundary in an image obtained through the camera, to an external display device through the communication module.

(115) According to various example embodiments, one or more of the at least one processor may be configured to: receive task area boundary information from a beacon device through the communication module, and set the task area boundary in the reference coordinate system based on the received task area boundary information.

(116) According to various example embodiments, the second marker image may be generated based on a drawing input related to the task area boundary, input to an external display device.

(117) According to various example embodiments, the second marker image may include a QR code image or a barcode image.

(118) According to various example embodiments, a method for controlling an electronic device may comprise: upon receiving a task performance command, moving to a task performing-related position identified based on the task performance command, obtaining a first marker image through a camera based on moving to the position, setting a reference coordinate system based on the first marker image, upon obtaining a second marker image through the camera, setting a task area boundary in the reference coordinate system based on task area boundary information obtained through the second marker image, and controlling a task performing member to operate within the task area boundary.

(119) According to various example embodiments, the task area boundary information may include at least one of coordinate information about at least one boundary surface, a slope, a shape of the task area boundary, or a minimum or maximum moving angle for each axis of the task performing member.

(120) According to various example embodiments, the task area boundary information may include at least one of coordinate information about a plurality of boundary surfaces or a slope. Setting the task area boundary may include obtaining information about at least one crossing line where the plurality of boundary surfaces cross each other based on at least one of the coordinate information about the plurality of boundary surfaces or the slope and setting the task area boundary based on the coordinate information about the plurality of boundary surfaces, the slope, and the information about the crossing line.

(121) According to various example embodiments, the method may further comprise, based on setting the reference coordinate system failing, reobtaining the first marker image through the camera.

(122) According to various example embodiments, the method may further comprise, based on setting the task area boundary failing, reobtaining the second marker image through the camera.

(123) According to various example embodiments, the method may further comprise transmitting an image obtained by displaying, in augmented reality, the task area boundary in an image obtained through the camera, to an external display device.

(124) According to various example embodiments, the method may further comprise receiving task area boundary information from a beacon device and setting the task area boundary in the reference coordinate system based on the received task area boundary information.

(125) According to various example embodiments, the second marker image may be generated based on a drawing input related to the task area boundary, input to an external display device.

(126) According to various example embodiments, the second marker image may include a QR code image or a barcode image.

(127) According to various example embodiments, a display device (e.g., the electronic device **101** of FIG. 1 or the display device **900** of FIG. 9) may comprise: a camera (e.g., the camera module **180** of FIG. 1), a display (e.g., the display module **160** of FIG. 1), and at least one processor (e.g., the processor **120** of FIG. 1) operatively connected to the camera and the display. One or more of the at least one processor may display, on the display, an image for a task space including a first marker image through the camera, receive a drawing input related to a task area boundary in the task space displayed on the display, display, in augmented reality, the drawing input in the task space displayed on the display, generate task area boundary information based on the user's drawing input on a reference coordinate system set based on the first marker image, and generate a second marker image including the task area boundary information.

(128) According to various example embodiments, the task area boundary information may include at least one of coordinate information about at least one boundary surface, a slope, a shape of the

task area boundary, or a minimum or maximum moving angle for each axis of the task performing member.

(129) The electronic device according to various embodiments of the disclosure may be one of various types of electronic devices. The electronic devices may include, for example, a portable communication device (e.g., a smartphone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, a home appliance, or the like. According to an embodiment of the disclosure, the electronic devices are not limited to those described above.

(130) It should be appreciated that various embodiments of the present disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be used to refer to similar or related elements. It is to be understood that a singular form of a noun corresponding to an item may include one or more of the things, unless the relevant context clearly indicates otherwise. As used herein, each of such phrases as “A or B,” “at least one of A and B,” “at least one of A or B,” “A, B, or C,” “at least one of A, B, and C,” and “at least one of A, B, or C,” may include all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as “1st” and “2nd,” or “first” and “second” may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively”, as “coupled with,” “coupled to,” “connected with,” or “connected to” another element (e.g., a second element), the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via a third element.

(131) As used herein, the term “module” may include a unit implemented in hardware, software, or firmware, or any combination thereof, and may interchangeably be used with other terms, for example, “logic,” “logic block,” “part,” or “circuitry”. A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, according to an embodiment, the module may be implemented in a form of an application-specific integrated circuit (ASIC).

(132) Various embodiments as set forth herein may be implemented as software (e.g., the program **140**) including one or more instructions that are stored in a storage medium (e.g., internal memory **136** or external memory **138**) that is readable by a machine (e.g., the electronic device **101**). For example, a processor (e.g., the processor **120**) of the machine (e.g., the electronic device **101**) may invoke at least one of the one or more instructions stored in the storage medium, and execute it, with or without using one or more other components under the control of the processor. This allows the machine to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include a code generated by a compiler or a code executable by an interpreter. The storage medium readable by the machine may be provided in the form of a non-transitory storage medium. Wherein, the “non-transitory” storage medium is a tangible device, and may not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between where data is semi-permanently stored in the storage medium and where the data is temporarily stored in the storage medium.

(133) According to an embodiment, a method according to various embodiments of the disclosure may be included and provided in a computer program product. The computer program products may be traded as commodities between sellers and buyers. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., compact disc read only memory (CD-ROM)), or be distributed (e.g., downloaded or uploaded) online via an application store (e.g., Play Store™), or between two user devices (e.g., smart phones) directly. If distributed online, at least part of the computer program product may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as memory of the manufacturer's

server, a server of the application store, or a relay server.

(134) According to various embodiments, each component (e.g., a module or a program) of the above-described components may include a single entity or multiple entities. Some of the plurality of entities may be separately disposed in different components. According to various embodiments, one or more of the above-described components may be omitted, or one or more other components may be added. Alternatively or additionally, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, according to various embodiments, the integrated component may still perform one or more functions of each of the plurality of components in the same or similar manner as they are performed by a corresponding one of the plurality of components before the integration. According to various embodiments, operations performed by the module, the program, or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.

(135) While the disclosure has been illustrated and described with reference to various example embodiments, it will be understood that the various example embodiments are intended to be illustrative, not limiting. It will be further understood by those skilled in the art that various changes in form and detail may be made without departing from the true spirit and full scope of the disclosure, including the appended claims and their equivalents. It will also be understood that any of the embodiment(s) described herein may be used in conjunction with any other embodiment(s) described herein.

Claims

1. An electronic device, comprising: a moving member comprising a motor; a task performing member comprising an arm; a communication module comprising communication circuitry; a camera; and at least one processor operatively connected to the moving member, the task performing member, the communication module, and the camera, wherein one or more of the at least one processor is configured to: based on receiving a task performance command through the communication module, control the moving member to move the electronic device to a position related to a task performing identified based on the task performance command; obtain a first marker image through the camera based on moving to the position; set a reference coordinate system based on the first marker image; based on obtaining a second marker image through the camera, set a task area boundary in the reference coordinate system based on task area boundary information obtained through the second marker image; and control at least one task performing member to operate within the task area boundary.
2. The electronic device of claim 1, wherein the task area boundary information includes at least one of coordinate information and a slope of at least one boundary surface, a shape of the task area boundary, or a minimum or maximum moving angle for each axis of the task performing member.
3. The electronic device of claim 1, wherein the task area boundary information includes at least one of coordinate information or slopes of a plurality of boundary surfaces, and wherein one or more of the at least one processor is configured to: obtain information about at least one crossing line where the plurality of boundary surfaces cross each other based on at least one of the coordinate information or the slopes of the plurality of boundary surfaces; and set the task area boundary based on the coordinate information and the slopes of the plurality of boundary surfaces, and the information about the crossing line.
4. The electronic device of claim 1, wherein one or more of the at least one processor is configured to, based on setting the reference coordinate system failing, reobtain the first marker image through the camera.
5. The electronic device of claim 1, wherein one or more of the at least one processor is configured to, based on setting the task area boundary failing, reobtain the second marker image through the

camera.

6. The electronic device of claim 1, wherein one or more of the at least one processor is configured to transmit an image in which the task area boundary is displayed in augmented reality on an image obtained through the camera, to an external display device through the communication module.

7. The electronic device of claim 1, wherein one or more of the at least one processor is configured to: receive task area boundary information from a beacon device through the communication module; and set the task area boundary in the reference coordinate system based on the received task area boundary information.

8. The electronic device of claim 1, wherein the second marker image is generated based on a drawing input related to the task area boundary, input to an external display device.

9. The electronic device of claim 1, wherein the second marker image includes a QR code image or a barcode image.

10. A method for controlling an electronic device, the method comprising: based on receiving a task performance command, moving the electronic device to a position related to a task performing identified based on the task performance command; obtaining a first marker image through a camera based on moving to the position; setting a reference coordinate system based on the first marker image; based on obtaining a second marker image through the camera, setting a task area boundary in the reference coordinate system based on task area boundary information obtained through the second marker image; and controlling a task performing member comprising an arm to operate within the task area boundary.

11. The method of claim 10, wherein the task area boundary information includes at least one of coordinate information and a slope of at least one boundary surface, a shape of the task area boundary, or a minimum or maximum moving angle for each axis of the task performing member.

12. The method of claim 10, wherein the task area boundary information includes at least one of coordinate information or slopes of a plurality of boundary surfaces, and wherein setting the task area boundary includes: obtaining information about at least one crossing line where the plurality of boundary surfaces cross each other based on at least one of the coordinate information or the slopes about the plurality of boundary surfaces; and setting the task area boundary based on the coordinate information and the slopes of the plurality of boundary surfaces, and the information about the crossing line.

13. The method of claim 10, further comprising, based on setting the reference coordinate system failing, reobtaining the first marker image through the camera.

14. The method of claim 10, further comprising, based on setting the task area boundary failing, reobtaining the second marker image through the camera.

15. The method of claim 10, further comprising transmitting an image in which the task area boundary is displayed in augmented reality on an image obtained through the camera, to an external display device.

16. The method of claim 10, further comprising: receiving task area boundary information from a beacon device through the communication module; and setting the task area boundary in the reference coordinate system based on the received task area boundary information.

17. The method of claim 10, wherein the second marker image is generated based on a drawing input related to the task area boundary, input to an external display device.

18. The method of claim 10, wherein the second marker image includes a QR code image or a barcode image.

19. A display device, comprising: a camera; a display; and at least one processor operatively connected to the camera and the display, wherein one or more of the at least one processor is configured to: display, on the display, an image, obtained by the camera, of a task space including a first marker image, receive a drawing input related to a task area boundary in the task space displayed on the display, display the drawing input in augmented reality in the task space displayed on the display, generate task area boundary information based on the drawing input in a reference

coordinate system set based on the first marker image, and generate a second marker image including the task area boundary information.

20. The display device of claim 19, wherein the task area boundary information includes at least one of coordinate information and a slope of at least one boundary surface, a shape of the task area boundary, or a minimum or maximum moving angle for each axis of the task performing member.
